Commons Bonds

Chapter 6: Three Ways of Counting

If cost severance is real — if market prices systematically understate the true cost of extracting from the commons — then how large is the gap? The previous chapters have shown, case by case, that the gap exists. Appalachian coal. Gulf oil. Libby vermiculite. A Chesapeake that is not the water it once was. Each is an illustration of the mechanism. None, by itself, answers the question a serious reader is entitled to ask: how do we know the cases are representative? How do we know we are not simply selecting disasters?

The answer matters because the framework's credibility depends on it. A mechanism demonstrated once is an anecdote. A mechanism demonstrated across cases is a pattern. A mechanism whose price can be computed, by multiple independent methods, and whose direction never flips across them, is a structural claim about how the economy works.

This chapter does the computing. Three approaches, built on different foundations, applied to the same historical cases. If the approaches converge — if three different accounting methods produce estimates in the same range — then the gap is not a methodological artifact. It is a measurable feature of the extraction economy. And the chapter closes by acknowledging, plainly, what the computation does not resolve. The carbon term dominates the fossil-fuel numbers and is politically volatile. The substitutability function requires engineering estimates that are themselves uncertain. The honest answer is not that the model eliminates uncertainty. The honest answer is that it structures uncertainty — tells us which parameters matter, and by how much each one affects what we are willing to call the price.

The three approaches are bottom-up damage accounting, a real-options model, and the residual commons value (RCV) framework. I will introduce each, apply each to McDowell County coal and to several other cases we have already met, and then compare their outputs in a single table. The reader who cares about the arithmetic can follow the arithmetic. The reader who cares about the argument can skip the formulas and follow the result.

Approach 1 — Bottom-Up Damage Accounting

Start with the method that needs no theory. Take one ton of Appalachian coal. Walk through every cost category the extraction produces. Enumerate them. Price each. Add.

Begin with health. The federal Black Lung Benefits Program's approximately $44 billion in distributions through 2009 (GAO/CRS), allocated across the national tonnage of coal produced since the Program's 1969 establishment, runs roughly eighty cents per ton (Chapter 2 walks the program's Trust Fund — its funding gap, debt trajectory, and structural mechanics — in detail). Add Medicare and Medicaid costs shifted to the public ledger by miners whose employers bankrupted before obligations matured. Add excess cardiovascular mortality from particulate exposure in communities downwind of coal plants. Add the respiratory illness in children who were never in the mine. A conservative national figure sits between $2 and $4 per ton.

Walk outside. Price the creek running orange with acid mine drainage. Price the tailings ponds whose liners were advertised for a century and are already failing at forty years. Price the headwater streams buried under mountaintop-removal overburden. Price the reclamation bond that was supposed to cover the closure — and note, as Chapter 2 did, that across Appalachia those bonds fall between \$3.7 and \$6 billion short of the work actually required. Environmental remediation, allocated nationally, runs another \$1 to \$3 per ton.

Price the communities. Not the extraction jobs — those went onto the benefit side of the ledger when the mine was operating. Price what remained. Price the outmigration that stripped the tax base. Price the schools that closed because the children left. Price the drug-epidemic costs that mining communities have borne at rates the rest of the country does not match. Price the life-expectancy gap — thirteen years, in McDowell County, against the national average. The thirteen-year gap is a *legacy effect* — a measurable consequence of cost-severance harm that persists across cohorts. The methodology for pricing legacy effects of this kind has been developed most rigorously by William Darity Jr. and collaborators in their work on the six-to-seven year Black-versus-white longevity gap in the United States, which prices the gap as a legacy effect of structural extraction across generations. McDowell County's thirteen-year gap is a legacy-effect pricing application of the same methodology, calibrated to coal-extraction-affected populations rather than to racially-stratified populations. The methodology travels; the per-context calibration is what differs. A conservative community-cost estimate comes in between \$5 and \$15 per ton.

Add them. Health yields \$2–4. Environment yields \$1–3. Community costs yield \$5–15. A conservative bottom-up total, absent carbon, is \$8 to \$22 per ton. Against a 1960 mine-mouth price of roughly \$4.50 and a contemporary mine-mouth price of \$40 to \$140 depending on region and grade, even without climate accounting the market has been capturing somewhere between roughly a quarter and a half of the actual cost. Not a rounding error. A factor.

Then add the carbon. This is where the arithmetic stops being conservative and starts being contested. One ton of bituminous coal, fully combusted, releases approximately 2.32 tons of CO_2 (EPA emission factor for bituminous coal at short-ton accounting: 93.28 kg CO_2 per mmBtu × roughly 24.9 mmBtu per short ton). At the EPA's 2023 estimate of \$190 per ton CO_2, the carbon externality alone is approximately \$441 per ton of coal — an order of magnitude larger than every other tier combined.

Include carbon, and the bottom-up total rises to roughly \$449 to \$464 per ton. Against \$4.50, or even \$140, the market is capturing roughly one to thirty percent of the actual cost. The carbon term dominates.

Apply the same exercise to other cases. Deepwater Horizon: production revenue of roughly \$4.25 billion against documented litigation costs, restoration obligations, ecological losses, and public-health spending in the range of \$65 to \$70 billion (a one-event disaster, so the cost figure is bounded by what has been counted rather than by an ongoing extraction lifetime). Call this ratio of true cost to market price the **Intergenerational Pricing Gap (IPG)**: 15 to 17 times for Deepwater Horizon. Libby, Montana: around \$100 million in lifetime mine revenue against \$5 to \$8 billion in direct settlements (approximately \$0.3 billion), federal cleanup spending (approximately \$0.6 billion and counting), and the long-tail of asbestos-related illness costs across thousands of cases, environmental remediation extending into future decades, and community destruction. An IPG of 55 to 82 times. Exxon Valdez: \$5.5 million of oil spilled (one-event disaster; the figure is product value of the spilled oil rather than an extraction-lifetime revenue stream), against between \$7 and \$10 billion in cleanup, litigation, and long-run ecological damage. An IPG in the range of 1,200 to 1,900 times.

Bottom-up accounting is a blunt instrument. It only captures the costs somebody has thought to count. It cannot price what no one has yet noticed. It is the accounting equivalent of a flashlight in a warehouse — what it illuminates is real, and what it misses is whatever is dark. And its quantitative output for fossil fuels lives or dies by the social cost of carbon, which lives or dies by whichever administration happens to be in office. The Obama EPA said $42 per ton CO_2. The first Trump administration said $1 to $7. The Biden EPA said $190. The same molecule. The same atmosphere. The same physics. Three estimates spanning two orders of magnitude.

The fight over the social cost of carbon (SCC) is itself a proxy war over acceptable cost severance levels. Every dollar of SCC that is not priced is a dollar of cost severed onto communities and future generations. The debate over the "correct" SCC is a debate over how much severance we are willing to tolerate.

The qualitative result — that the market underprices extraction — is robust across all SCC estimates above zero. The quantitative result is not. The framework's numbers for fossil fuels are only as precise as the SCC estimate chosen, and the SCC estimate is, at the moment of this writing, whatever the sitting administration says it is. The framework does not deliver a definitive number for the true cost of a ton of coal. It delivers a range, and the range depends on parameters whose estimation is itself a contested political and scientific enterprise. What the framework does is make the dependency explicit. It tells the reader which parameter moves the output by how much and invites the reader to judge, rather than hiding the judgment behind a single headline number.

The choice of $3 per ton is not a conservative estimate in any epistemic sense. It is a political statement — an announcement that the extractor should pay very little for the carbon released. The choice of $190 is a different political statement, informed by a larger body of climate science, but still a choice. There is no price-free default here. The available alternative — current market pricing with no externality adjustment at all — is not neutral either. It is a specific assumption: that externalities are worth exactly zero. That the health costs borne by McDowell County are worth zero. That the climate damage from atmospheric carbon is worth zero. That permanent foreclosure of a non-renewable resource imposes zero cost on future generations. This assumption is not more rigorous than the framework's. It is less rigorous — it is an assumption that contradicts every piece of available evidence.

The framework does not eliminate uncertainty; it structures uncertainty. It tells the reader which parameters matter and by how much each one affects the output. A weather model does not tell you with certainty whether it will rain Thursday; it tells you the probability, given what we know. The weather model is not useless because it contains uncertainty. It is useful because it expresses the uncertainty in a form that supports decisions. The same logic applies here. And the parameters are becoming more estimable, not less. The social cost of carbon was genuinely uncertain in 2010. By 2023, the EPA's $190/ton estimate reflected vastly more sophisticated climate modeling, economic damage functions, and discount-rate analysis than any earlier estimate. Substitutability curves for specific resources are being refined by energy transition modelers every year. The uncertainty range is narrowing. A framework that structures ongoing research is more useful than one that ignores the problem because precise answers are not yet available.

One distinction worth making explicit at this point in the carbon accounting. The bottom-up damage-accounting we have been doing prices what carbon *does* once emitted — the damage side, anchored by the Social Cost of Carbon. The framework's forward-looking residual commons value apparatus (Approach 3, below) also asks a different question — what would it cost to engineer the substitute the commons was providing? — and the answer for atmospheric CO_2 comes from direct-air-capture. Both numbers belong in honest carbon accounting; they answer different questions.

Climeworks's Orca facility in Iceland, currently among the largest operational direct-air-capture installations, captures CO_2 at approximately one thousand dollars or more per ton at first-of-a-kind operational scale. Climeworks's Mammoth facility (operational from 2024) and the company's Generation-3 technology trajectory project unit cost reductions toward four hundred to six hundred dollars per ton net-removal cost by 2030, under the corporate cost-reduction trajectory the company disclosed at its June 2024 Carbon Removal Summit in Zurich. The capture-only target along the same trajectory is two hundred fifty to three hundred fifty dollars per ton. Carbon Engineering's Stratos facility in Texas, in commissioning at the time of writing with phased ramp to full capacity through mid-2026, is projected to capture at three hundred to six hundred dollars per ton at full operation. The International Energy Agency's *Direct Air Capture 2022* review places the achievable cost at scaled-up operation in the two-hundred-thirty to six-hundred-dollar range, depending on energy-source assumptions. The IPCC's AR6 Working Group III, under more optimistic learning-curve and policy-support scenarios, projects mid-century costs as low as one hundred to three hundred dollars per ton. The book presents all four horizons because the answer depends on which one the reader is asking about, and the framework's response to the question "what does it cost to replace what's been severed" is honestly horizon-dependent.

The Social Cost of Carbon — the figure of approximately one hundred ninety dollars per ton at a two percent discount rate, drawn from Rennert and colleagues 2022 — prices the *damage* that a ton of CO_2 does once emitted; direct-air-capture prices the *engineering reversal* of that damage. The substitution-side question (what does it cost to replace the commons function) belongs in Approach 3's RCV Method 1; the damage-side question (what does the harm cost) is what Approach 1's bottom-up walkthrough has been answering. Conflating them is one of the easier methodological errors a reader could make, and the framework's reporting discipline keeps them separated so the reader doesn't.

Even with this limitation, the bottom-up method gives a floor. It says that the market underprices extraction by at least a factor of several, and potentially by factors of tens or hundreds. It does not tell us the ceiling. For that, we need a method that does not rely on enumerating costs one by one.

Approach 2 — The Real Options Model

The second method speaks the language of Wall Street.

A barrel of oil in the ground is not merely a stock of energy. It is an option — the right, but not the obligation, to extract at any future time. Options have value. The longer you wait, the more information you have about what the resource will be worth, what substitutes will exist, and what externality costs will prove to be. Premature exercise of an option destroys the information value that waiting would have produced.

This is standard finance. It is how every oil major already values its undeveloped reserves internally. The model is not radical. It is the mainstream model.

What the standard model omits is the social side of the ledger. The private option value — what a resource owner gains by waiting — is only a fraction of the total option value. The social option value also includes the externality costs avoided by extracting later (better remediation technology, better substitutes, smaller externality tail); the substitutability that develops while you wait (solar cells at three cents per kilowatt-hour that did not exist in 1960); and the information about resource criticality that accumulates over decades (we did not know, in 1960, that certain rare earth elements would be required for every piece of modern electronics).

When the full social option value of an unextracted barrel exceeds the net revenue from extracting it today, extraction is socially premature. Not suboptimal. Premature. The economy is exercising an option before the information the option contains has been collected.

Apply this to McDowell County coal in 1960. The mine-mouth price was roughly $4.50. Net revenue per ton, after extraction costs, was well under that. The social option value of the same ton — priced by a later substitutability window (renewables by the 2020s were already cheaper than coal in most markets), avoided externalities, and the information that would have been collected about Appalachian community collapse — is not $4.50. It is on the order of tens of dollars at minimum, and plausibly hundreds once the externality tail is included.

1960 coal was socially premature. The option should not have been exercised. That we exercised it anyway is not a failure of information — much of the information was suppressed deliberately — but a failure of pricing. The market gave the option to the extractor at no cost. The social side of the ledger was not a line item.

The real-options model is useful because it speaks to an audience the damage-function approach does not reach. A CFO who would dismiss a bottom-up list of externalities as bleeding-heart arithmetic will recognize option value as a real quantity. She values her own company's reserves this way every quarter. The only change is the addition of the social terms.

Apply it across cases, and the same answer emerges. Every extraction event we have examined produced net revenue substantially below the social option value of leaving the resource in place. The direction does not flip.

Approach 3 — The RCV Model

The third approach is the one the book has been building toward, and it deserves careful introduction because its architecture is different from the first two. The bottom-up method enumerates costs. The real-options method prices what is lost by exercising early. The residual commons value model prices something neither of the first two capture: what every future generation loses by not having the resource, weighted by the probability that no substitute will exist by the time they need it.

I will introduce the formula in a moment. Before the formula, a plain-English version of what the formula computes, because the argument has to survive without the mathematics for readers who would rather not do integrals.

The true cost of extracting from the commons — most dramatically for a non-renewable resource, and, with adjustment, for a renewable one extracted at rates exceeding regeneration — has two components the market does not currently price.

The first is the foreclosure cost. When you extract a barrel of oil, every future generation loses access to that barrel. If a substitute exists — if renewable energy can do everything oil does, for every application oil is used for — then the foreclosure does not matter. Future generations do not care about the oil because they have something better. But if no substitute exists for certain applications — if you need physical petroleum for specific materials, specific chemicals, certain industrial processes that nothing else can replace — then the foreclosure is real. The future generation that needs the oil and cannot get it bears a genuine cost.

Foreclosure cost depends on substitutability: how quickly, how completely, and at what cost alternatives can replace the resource. For coal in electricity generation, substitutability is high and rising fast — renewables are already displacing coal at scale. For rare earth elements in advanced electronics and propulsion, substitutability is low — no alternative provides the same magnetic, catalytic, or luminescent properties, and materials scientists have been looking for decades. For helium, substitutability in cryogenic applications may be effectively zero. For fossil aquifer water, substitutability is zero on any timescale that matters to the people drinking it.

The second component is the externality tail. When coal is mined, the health damage does not stop when the mine closes. Black lung kills miners decades after their last shift. Acid mine drainage runs for generations. Carbon stays in the atmosphere for centuries. These ongoing costs are real, measurable, and currently unpriced by markets. A substitute for coal does not clean up the mine. A substitute for oil does not remove carbon from the atmosphere. The externality tail runs independently of substitutability.

The framework's choice of *externality tail* as the name for the long-running damage that cost severance produces is a borrowed metaphor from statistics, and the borrowing is deliberate. In probability theory, the tail of a distribution is the region where probability mass survives at extreme values — the part of the curve that does not collapse to zero as the variable runs to its limits. Heavy-tailed distributions are those whose expected value is dominated by the tail rather than by the body; the rare-but-large events drive the average, and analysts who price only the body of the distribution systematically under-estimate the expected cost. The framework borrows the precise sense. Carbon released by burning Norwegian oil today will remain in the atmosphere for centuries; acid mine drainage from Appalachian coal seams runs for generations; the asbestos-related illnesses from Libby's vermiculite will kill people whose grandparents were not yet born when the mine closed. These costs do not have an end-date the way conventional externality accounting presumes. They continue indefinitely with non-vanishing magnitude, and the cumulative cost is dominated by the long-running tail rather than by the immediate damage. Other names were available and each fails for a specific reason. *Externality aftermath* implies a bounded period after the event, which is structurally wrong — the damage does not have an aftermath in the sense an event has an aftermath; it has a tail in the sense a distribution has a tail. *Externality trail* implies a path that could be cleaned, which the framework specifically rejects (no fund extracts carbon already in the sky). Three weaker alternatives also fail: *externality legacy* implies a named-recipient obligation the framework handles separately at the bond-instrument level; *long-run externality* loses the distribution-shape claim (long-run could be bounded; the tail is not); *persistent externality* names the duration without naming the shape. The framework's tail is in the statistical-distribution lineage that climate-tipping-point and climate-tail-risk literatures occupy; Weitzman's 2009 work on fat-tail climate damages is the adjacent tradition the usage joins. Naming the tail is what makes the bond's forward-posting architecture necessary: a bond posted against bounded aftermath could discharge when the aftermath ends; a bond posted against a tail must carry forward indefinitely, because the tail does not end on its own.

The residual commons value combines these two components: foreclosure cost, weighted by the probability that no substitute exists at each future time, plus the externality tail, both integrated across the full time horizon affected by the extraction. When the RCV exceeds the market price — which, as I will show, it does in every case we have tested — the difference is the cost that has been severed.

Now the formula — a summary of the plain-English argument just made:

$$\mathrm{RCV}(R, t_0) = \int_{t_0}^{\infty} \{[1 - S(t \mid t_0)] \cdot U(R, t, Q(t)) + E(R, t)\} \cdot D(t, t_0)\, dt$$

(formal articulation in Technical Appendix §3 — RCV Quantification — Three Ways of Counting)

The integral sign announces that we are summing contributions across all future time. The bracketed expression inside is the per-period cost: what is lost in year t, discounted back to year t_0, weighted by how likely it is that anything in that year can substitute for the lost resource.

Translate term by term:

- $[1 - S(t \mid t_0)]$ is the probability that no adequate substitute exists at time t, given that extraction occurred at t_0. If S = 1 (perfect substitute exists), this term is zero and the foreclosure cost vanishes for that period. If S = 0 (no substitute), this term is 1 and the full foreclosure cost counts.
- $U(R, t, Q(t))$ is the utility future generation t derives from having one unit of R available in situ, given the remaining stock Q at time t. As stock depletes, U rises because each remaining unit becomes more valuable.
- $E(R, t)$ is the externality flow at time t: ongoing health damage, remediation costs, climate damage, community disruption. This term runs independently of substitutability and independently of the resource having been extracted to depletion.
- $D(t, t_0)$ is the social discount factor, following Weitzman's declining-rate framework: under uncertainty about the long-horizon discount rate, the certainty-equivalent rate declines toward the lowest plausible rate. This is a technical move but a principled one: it converts an unresolvable philosophical question (what is the correct discount rate?) into a treatable empirical one (given uncertainty, what is the certainty-equivalent?).

Cost severance is what is left after you subtract from RCV whatever accountability bond B has actually been posted:

$$\mathrm{CS} = \mathrm{RCV} - B$$

And the intergenerational pricing gap is the ratio of true cost to market price:

$$\mathrm{IPG} = \mathrm{RCV}/P_{\text{market}}$$

That is the entire mathematical apparatus this chapter requires. The appendix derives convergence conditions, proves that the integral is finite under reasonable assumptions, and demonstrates that for non-renewable resources under any existing accountability regime, CS is strictly positive. The reader who wants the proofs will find them there. The reader who wants the argument already has it.

The Three Ways of Counting Inside RCV

Inside the residual commons value computation, a triangulation operates that the formula above does not display on its face. The RCV estimate is not produced by a single estimation method; it is produced by three. The reason has to do with an institutional asymmetry financial economics is well-positioned to recognize. Markets do have a mechanism for pricing future profitability — discounted-cash-flow analysis aggregated through equity and debt markets, embedded in every asset price as an implicit expectation about future returns. The mechanism is implicit but real. Markets do not have an equivalent mechanism for pricing future harms. No aggregating institution produces a comparable price signal for the externality tail, for foreclosure cost, or for the long-tail damage of cost severance. The three estimation methods — Replacement Cost, Revealed Preference, Scarcity-Adjusted Option Value — are the framework's substitute for the missing market mechanism on the harms side. They give the harm side a comparable price signal to the one the asset side already has, so that the asymmetry between priced future profitability and unpriced future harms is no longer the institutional default. Each method anchors its estimate in a different empirical pathway; the convergence range across the three is what the framework reports as the RCV estimate. The triangulation is the work that gives the RCV claim its standing.

Method 1 — Replacement Cost. The first method asks: what would it cost to engineer a substitute for the commons function that extraction is consuming? Applied to McDowell coal through its carbon dimension, the substitute for atmospheric CO_2-removal is direct air capture — engineered facilities that pull CO_2 out of the air at the per-ton-of-CO_2 costs documented above. Translated to a per-ton-of-coal figure under the bituminous-coal emission factor and across DAC's documented horizons, Method 1 lands in the three-hundred-ten to eighteen-hundred-dollar-per-ton-of-coal range. This is the most directly priced of the three methods and the smallest epistemic-ambition step: the substitute is buildable, the cost is engineered, the floor is real. The limit is that Method 1 anchors only the substitutable component of the commons function; it underprices irreplaceable commons. Direct air capture engineers atmospheric CO_2 removal but does not replace foreclosed petroleum hydrocarbons or restore destroyed communities, and Method 1 is treated as a floor for that reason. (Formal specification in Technical Appendix §3.3 — Method 1: Replacement Cost.)

Method 2 — Revealed Preference. The second method asks: what does a polity that explicitly chose to capture the rent from its extraction and pass it forward — rather than allow the rent to be consumed at extraction-time — reveal about commons-value through its observed restraint? Norway is the primary anchor for the method: the Government Pension Fund Global is a polity's revealed valuation of forward foreclosure-prevention, observed at scale, with a documented per-barrel-of-oil-equivalent figure of approximately fifty dollars. The method does not translate Norway's per-barrel figure arithmetically to other extractions; the method extracts what Norway has revealed about polity-level willingness to pay for foreclosure-prevention and applies that revealed-preference reasoning to other extraction contexts. Applied to McDowell coal under comparable revealed-preference reasoning, Method 2 lands in the eight-to-eighty-eight-dollar-per-ton range. This is the empirically-grounded middle of the triangulation: a real polity has actually paid for a real instance of forward foreclosure-prevention, and the figure does not rely on engineering counterfactuals or option-value assumptions. The limit is that Norway's capture is per-Norwegian-citizen accounting; the atmospheric externality non-Norwegian populations bear from Norwegian oil is not in the Government Pension Fund Global. Method 2 sits as a revealed-preference-anchored middle of the triangulation but as a lower bound on true RCV for that reason. (Formal specification in Technical Appendix §3.4 — Method 2: Revealed Preference.)

Method 3 — Scarcity-Adjusted Option Value. The third method asks: what is the option value of preserving the commons under explicit scarcity-and-irreversibility weighting? Applied to McDowell coal via Dixit and Pindyck's 1994 real-options framework with the framework's scarcity multiplier (logarithmic in commons-stock-to-sustainable-flow ratio) and irreversibility premium (calibrated to the probability the commons cannot be restored at any finite cost once extracted), Method 3 produces a per-ton range of four hundred twenty to thirteen thousand one hundred dollars, with a mid-range estimate around twenty-five hundred dollars per ton. For regimes where coal extraction operates as a commons-extinction event, the irreversibility parameter sits in the 0.9-to-1.0 range — the α-dominance regime, in which the irreversibility premium dominates the scarcity multiplier — per the Technical Appendix's cross-case sensitivity analysis. This is the framework's upper bound: a method that prices what extraction forecloses under explicit recognition that the foreclosed commons cannot be brought back at any finite cost. The wide numerical range reflects scarcity-and-irreversibility parameter uncertainty rather than methodological imprecision; the α-dominance finding across the documented case library is the empirical signal that the asymmetric-regret structure is not an edge case but the dominant cross-case driver of commons-extraction-cost variance. (Formal specification in Technical Appendix §3.5 — Method 3: Scarcity-Adjusted Option Value.)

The triangulation, applied forward and backward. The three methods together — Replacement Cost as floor, Revealed Preference as middle, Scarcity-Adjusted Option Value as ceiling — produce a convergence range rather than a point estimate. The framework's reporting discipline requires that every RCV estimate publish all three method figures plus the convergence range plus identified divergence sources; no single-method RCV claim has standing. The same triangulation runs in reverse. Where extraction has already occurred — Chapter 5's backward-direction application of the apparatus to legacy effects already realized — Method 1 — Replacement Cost specializes to remediation cost rather than substitute-engineering cost; Method 2 — Revealed Preference specializes to revealed restitution (the case lineage of Holocaust reparations, the 1988 Civil Liberties Act for Japanese-American internment redress, Black Lung Trust Fund payouts, the South African Truth and Reconciliation Commission, and the reparations-economics methodology Darity and Mullen 2020 developed for the racial-wealth-gap accumulated through American policy choices since 1619); Method 3 — Scarcity-Adjusted Option Value specializes to the option value extinguished at the time of past extraction, evaluable retrospectively from parameters known at analysis time. The methodology is direction-agnostic; the calibration adjusts per case. (The bidirectional structural property is formalized in Technical Appendix §5.5.)

The Substitutability Function

Before turning to the convergence result, a word about what the substitutability function is doing, because it is the model's most novel move and also its most vulnerable.

Previous approaches to pricing intergenerational costs relied almost entirely on the discount rate. How much do future generations matter compared to this one? Economists have been arguing about the answer since Ramsey in 1928. They have not converged. The Stern Review used 1.4 percent. Nordhaus used roughly 4 percent. Weitzman argued for a declining rate. Philosophers have argued that any positive discount rate on well-being is unjustifiable. Different administrations have set the social cost of carbon at \$3, \$42, and \$190 across the last decade. The philosophical question refuses to be settled.

The substitutability function sidesteps this impasse. Instead of asking "how much do future generations matter?" it asks "how quickly can we develop alternatives?" This is an empirical question. Engineers and materials scientists can estimate substitution timelines. Energy economists can model the cost trajectory of renewable technologies. Hydrologists can estimate aquifer recharge rates. The question is technical, not philosophical. And technical questions admit of progress.

The substitutability function has four natural properties. It starts near zero at the moment of extraction — if a substitute already existed and was cheaper, we would not be extracting. It rises monotonically over time; substitutes, once invented, do not un-invent themselves. It approaches some asymptotic limit $S(\infty|t_0) \leq 1$; for some resources, complete substitution may never be physically possible. And it depends on investment; the timeline for a given substitute is not fixed but responsive to the price signals and research priorities that an honest pricing regime would generate.

The critic's objection is fair. The substitutability function is doing enormous work in the model, and in practice it must be estimated. A critic can say, correctly, that we have moved the hard problem from choosing a discount rate to choosing a substitutability curve. True. But it is a better hard problem — substitutability is an empirical question whose error bars shrink with research, whereas the discount rate is a philosophical one whose error bars have not shrunk in a century. (Formal specification in Technical Appendix §13 — Substitutability Function and Gap.)

The Convergence

The three approaches — bottom-up, real-options, and RCV — are built on different foundations. They use different mathematical structures. They make different assumptions about what to count and how to aggregate. If they all produced similar answers by accident, one would be entitled to suspicion. If they all produced similar answers because the underlying reality is real, one would expect exactly that.

A simpler framework would have offered a single estimation approach — bottom-up damage-accounting alone, or the real-options model alone, or the RCV model alone. A single approach would have been simpler to teach, simpler to implement, and simpler to defend. It would also have inherited that single approach's pathologies. Bottom-up damage-accounting is grounded in identified damages but cannot capture costs no one has yet thought to count. The real-options model prices substitutability under irreversibility but inherits the parameter-calibration uncertainty in the substitutability function. The RCV model integrates across future generations but depends on the stock-dependent utility and externality-tail estimates the analyst chooses. Each approach has a strength the other two lack and a blind spot the other two illuminate. Triangulation is the standard scientific-method response to estimation under uncertainty: produce three independent estimates from three independent epistemological angles, report the convergence range, treat divergence as informative rather than as failure. (The three-model convergence is formalized in Technical Appendix §9 — Three-Model Convergence.)

Apply the inner three methods — Replacement Cost, Revealed Preference, Scarcity-Adjusted Option Value — to the cases the framework has worked with, and compare:

Case	Market Price	M1 — Replacement Cost	M2 — Revealed Preference	M3 — Scarcity-Adjusted Option Value	Direction
McDowell County coal	$4.50/ton (1960 mine-mouth)	$261–$2,412/ton	$8–$88/ton	$420–$13,100/ton; mid $2,500	CS > 0
Norway petroleum	~$48/BOE rent captured	$161–$422/BOE	$48/BOE	$70–$1,000/BOE; mid $280	CS > 0
Deepwater Horizon	$4.25B Macondo revenue	~$22B engineering + ecological restoration (BP response ~$14B + NRDA $8.8B per 2016 Consent Decree)	~$20–$25B settlements + damages paid (CWA $5.5B + DOJ criminal $4.5B + class-action economic ~$10–$15B)	N/A†	CS > 0
Libby, Montana	~$100M lifetime revenue	~$0.6B EPA Superfund cleanup spending + ongoing remediation; long-tail illness-cost flow $4–$7B	~$0.3B W.R. Grace settlements ($250M 2008 EPA + $18.5M 2023 NRDA + $5.1M 2008 DEQ)	N/A†	CS > 0
Exxon Valdez	$5.5M product spilled	~$2.3B cleanup + ecological restoration (Exxon $2.2B + NRDA ~$100M per 1991 settlement)	~$1.9B settlements + damages paid (1991 civil $900M + private claims ~$507M + punitive $507M + criminal $25M)	N/A†	CS > 0

† Method 3 (Scarcity-Adjusted Option Value) does not cleanly apply to one-event historical cases at the forward direction: the option-value framing prices what extraction forecloses on future generations, and for completed historical events the forward-foreclosure question is moot. The backward-direction application of Method 3 (option value extinguished at the time of past extraction, evaluable retrospectively

from parameters known at analysis time, per Technical Appendix §5.5) is structurally available for these cases but sits outside this volume's empirical scope.

The M1, M2, and M3 cells report per-method estimates anchored to the method specifications in Technical Appendix §§3.3–3.5. McDowell County coal and Norway petroleum figures are sourced verbatim from the Technical Appendix's worked-example calibration (§3.6 Block 4 numerical execution). Deepwater Horizon, Libby, and Exxon Valdez figures are derived from the method specifications applied to each case's documented cost record, with cell values cross-referenced against authoritative public sources (NOAA Office of Response and Restoration, DOJ, and the 2016 BP Consent Decree for Deepwater; the EPA Libby Asbestos Superfund Site record, Montana Department of Justice, and W.R. Grace settlement public filings for Libby; the Exxon Valdez Oil Spill Trustee Council and DOJ 1991 settlement record for Exxon Valdez). A reader comparing the M1, M2, and M3 ranges across cases will see ranges rather than single numbers: the per-method convergence range is the framework's reporting discipline, and the divergences across methods are empirically informative — surfacing what feature of the commons each method captures.

Numerical validation across the case library cross-confirms the per-case settlement figures at the lower end of the triangulated three-method range.

McDowell coal's per-case implied IPG of approximately fifty times (per the Technical Appendix's implied cost-severance summary in §3.6 Block 4) falls inside the triangulated fifty-to-five-hundred-fifty-five-times range when the M1, M2, and M3 estimates are aggregated estimate-by-estimate; the variance reflects time-horizon attribution (1960 nominal mine-mouth price versus 2025 real cost-severance) more than framework imprecision.

Libby's documented cost-to-revenue ratio of forty-times sits inside the M1-and-M2-anchored cleanup-plus-settlements aggregate, with the long-tail illness-cost flow documented as a separate stream the M1 column captures but the per-case settlement work compresses to a single ratio.

Deepwater's forty-percent cost-recovery ratio (BP paid out roughly forty percent of total documented damages through the M2 settlement pathway; the remainder runs through the M1 restoration pathway plus uncounted long-tail damage) takes a different form than the multiplicative IPG figures because Deepwater is a one-event disaster — total documented cost is bounded — versus the extraction-lifetime cases, where IPG multiples capture cumulative underpricing across decades.

Every cost-severance figure carries an implicit method-and-time-horizon attribution, and the convergence-table presentation reports the range across methods rather than collapsing to a single number. Readers who want the simplest defensible single-number version per case will find it where the case is introduced; readers who want the full method-level decomposition will find it in the Technical Appendix. The convergence-table ranges sit between the two — broader than the per-case headlines, narrower than the full method-decomposition — and make the methodological structure visible in a single view.

The methods converge. Different assumptions, different foundations, same qualitative conclusion and, in most cases, similar magnitudes. The direction never flips. In every case tested, market prices understate the true cost — always by a meaningful factor, sometimes by three or four orders of magnitude.

This is the chapter's central finding, and it is worth sitting with for a moment. Three independent accounting methods, built on economics from different decades and different schools, have been applied to the same historical extractions. They agree not because they were designed to agree but because the thing they are measuring is real. The gap between what extraction costs and what extraction charges is not an artifact of any one method. It is a feature of how the extraction economy actually works.

Three more questions arise at this point, and the chapter owes the reader an answer to each.

The first is the "choose-your-own-adventure" objection: that the model's output depends on parameter choices the analyst makes, and that different choices produce different numbers. This is partly correct. But the convergence above is itself evidence against arbitrariness. If the output were purely an artifact of arbitrary parameter choices, three different methods built on different foundations would not converge. They do. The convergence is evidence that the models are measuring something real, however imprecisely.

The second is the Popperian falsifiability question. A model that says extraction is always underpriced, the philosopher of science will say, is unfalsifiable. The correct response is that the model is falsifiable — the book has simply not yet found a case where it fails. Recall the definition: $CS = RCV - B$. The model produces $CS > 0$ only when the accountability bond B is smaller than the true intergenerational cost. CS can theoretically equal zero or be negative. Two kinds of case approach this possibility. A sustainably managed renewable resource with full ecosystem accountability — timber from a plantation forest harvested at or below regrowth rates, with full replanting and habitat maintenance bonded to the operation — approaches $CS \approx 0$; the foreclosure component is zero because the resource regenerates, and the bond covers the remainder. A highly abundant non-renewable extracted under stringent accountability — common construction sand from a lawfully permitted pit in a jurisdiction with full reclamation bonding — approaches $CS \approx 0$ by a different path; the stock-dependent utility is near zero because the resource is effectively non-scarce on any meaningful timescale, and the bond can match the RCV. Both kinds of case are practically attainable; both predictions are testable; both would be falsified by clean counterexamples.

The stronger response is the structural one. The reason $CS > 0$ in every fossil-fuel case we have tested is not that the model is rigged. It is that no existing accountability regime incorporates substitutability-weighted intergenerational foreclosure pricing. The instruments do not yet exist. The model's prediction is that under current instruments, $CS > 0$ for non-renewables; that prediction will be falsified the day a jurisdiction implements full intergenerational foreclosure pricing and demonstrates $CS \leq 0$ for a non-renewable resource. This is analogous to the claim, made in 1940, that no existing airplane could fly faster than sound. The claim was true, not because it was unfalsifiable, but because the necessary technology had not been built. It was falsified in 1947 when Chuck Yeager flew through the barrier.

Two distinct falsifiability operations are at work. The Commons Inversion Test is a thought-experiment filter — a candidate cost claim either survives or fails its counterfactual inversion. *Would the cost be priced differently if the commons were absent? Would it be priced differently if the commons were inexhaustible?* This is conceptual filtering; thought-experiments don't survive empirical falsification in the Popperian sense, and the framework does not claim they do. The empirical falsifiability operates one layer above it, at the level of specific cost-claim predictions. *$CS > 0$ for case X under existing accountability regimes.* That prediction is empirically testable through the case-study record; it is what the three approaches calibrate. Until full intergenerational foreclosure pricing is built, the universal $CS > 0$ prediction stands, and its universality is evidence of a structural gap, not a methodological trick.

The third is the reproducibility question: two analysts applying the framework to the same case will not, in general, produce the same number. They will choose different substitutability curves, different social-cost-of-carbon estimates, different discount factors, different assumptions about the externality tail. Their outputs will overlap but will not be identical. This is how applied cost-accounting works across every framework in active use — benefit-cost analysis of environmental regulations, health-technology assessment, nuclear decommissioning cost estimates, actuarial forecasts. None of these fields treats the presence of a range as a failure of the method. What they treat as a failure is an estimate whose sensitivity to parameter choice is not disclosed, or whose central case is invented rather than derived from data. The framework guarantees not uniform outputs but transparent disagreement — outputs whose dependence on each parameter is exposed rather than hidden, and whose disagreements resolve to specific empirical questions that further research can answer.

The Norway Backtest

To test whether the model only screams disaster, apply it to the best-managed extraction case in the world.

Norwegian oil. The Norwegian sovereign wealth fund — formally, the Government Pension Fund Global — holds approximately $2.2 trillion (early 2025 NBIM) against the country's cumulative oil revenues and invests that capital in a diversified portfolio on behalf of present and future generations. It is the closest working example the world has of Hartwick's Rule: extract a non-renewable, invest the rents in renewable capital, let consumption depend on the interest rather than the principal.

Apply the residual commons value framework. The fund is an enormous accountability bond. The country has stringent environmental regulation and has invested in full-cost accounting for the externalities of its extraction industry. The externality tail is smaller than in Appalachia because the regulatory regime is stricter. The substitutability picture for oil is the same as everywhere else.

Result: CS is small, but positive. The fund captures most of the financial value. The barrel is still physically gone; no fund can manufacture more subsurface petroleum. The externality tail, though smaller, is not zero; Norwegian oil burned anywhere on Earth contributes to the same atmosphere. The residual commons value is not so much larger than the accountability bond, but it is still larger.

The model does what a good model should. It differentiates. It does not say "extraction is always bad at the same rate." It says "extraction under these conditions produces this much cost severance, and extraction under those conditions produces that much." Norway is closer to honest pricing than anywhere else. It is still not all the way there.

Ten Commons Categories — Examples From the Cases This Book Examines

Across the eighteen cases this book examines, the Commons Inversion Test has repeatedly surfaced ten commons categories worth naming as examples:

Habitability · Spatial · Temporal · Institutional · Kindred · Ecosystem · Political · Cohesion · Epistemic · Autonomy.

These are *examples for illustrative purposes, not an exhaustive list.* The ten represent what showed up across the eighteen cases tested; readers applying the methodology to their own cases may surface different categories or refine these. The framework's universality is about the method, not about whether ten is the right number. Ostrom listed eight design principles. Sen and Nussbaum elaborated ten central capabilities. Rawls listed primary goods. Each of those frameworks succeeded by being honest about what they had identified — examples drawn from cases, not metaphysical claims about what exists.

A note before going further. This framework doesn't argue that autonomy is universally a commons, or that habitability is, or that time is. It argues that when any of these operate *as* commons in a given society — shared, depended-on, collectively enabling — extraction can sever cost from that commons, and this methodology measures the severance.

Different political traditions have different commitments about what is shared and what is individual. Classical liberalism treats autonomy as individual natural-right; civic republicanism treats autonomy as non-domination requiring shared institutions; socialist and communitarian traditions treat autonomy as enabled by shared infrastructure; Marxist analysis treats autonomy as material-conditions dependent; lived-oppression perspectives treat autonomy as something extracted, not shared. The framework works within any of those commitments. What it doesn't accept is extraction without accountability, regardless of which commons is being extracted from.

The available alternative would have been legislative — declare these ten as the closed and complete enumeration of commons, and route any candidate eleventh through a framework-controlled admission process. The framework rejects that posture because what counts as a commons is politically and traditionally contested; a framework that legislates the ten as definitive declares one political-philosophical tradition's commitments correct and the others incorrect, and surfaces the framework's own political commitments as a precondition for using its accounting.

The ten categories above are examples drawn from cases this book examines. A reader whose tradition handles them differently can apply the same methodology and surface different categories. The throughline is the mechanism, not the enumeration.

Autonomy is the most contested of these examples, and worth lingering on. Classical liberals will recognize autonomy as individual natural-right; the supporting infrastructure (rule of law; contract enforceability; labor protections; severance norms) is enabling but not itself commons. Civic republicans, in the tradition of Anderson, Pettit, and Skinner, will recognize autonomy as non-domination, requiring shared institutions to remain sustainable — institutions that function as commons. The framework doesn't settle this debate. It accommodates both readings — and the lived-oppression reading, in which autonomy is something extracted from rather than shared with the bearer. What the framework asks: when extraction operates on whatever autonomy-supporting infrastructure your case treats as load-bearing, the cost-severance methodology applies. The Commons Inversion Test imagines the supporting infrastructure absent or consumed; the costs that become visible are commons-grounded; they admit to RCV.

Chapter 1's account of a knowledge-worker billing 120 hours per week is the framework's most-favorable-conditions test of this proposition. The billing was voluntary. The contract was negotiated. The worker had elite credentials and could have walked. And what the contract priced was the billable time, not the displacement — not the spouse recovering from dying, not the son who could have been held longer, not the autonomy-supporting infrastructure (employment law shaping severance norms; the social architecture that makes elite-knowledge-worker autonomy possible) that the contract architecture systematically eroded. Whether you read that as a Commons-Consumption Inversion against an autonomy commons (the civic-republican reading), as a market mechanism functioning as designed (the classical-liberal reading), or as evidence of structural domination (the lived-oppression reading), the framework's measurement applies. That is the framework's claim to universality: not that one tradition is right, but that the cost-severance methodology operates across them.

This chapter walks two of the ten commons categories at depth — habitability through the McDowell case, and time / autonomy through the personal-scale Consumption-Inversion examples carried over from Chapter 1. The remaining eight commons categories work through the same methodology applied to different cases. Their per-category specifications, with the full worked-example record from the eighteen cases this book has examined, are catalogued in the Technical Appendix. Readers who want to see the methodology applied across the full case library will find it there. Readers applying the framework to their own cases will find their own commons categories — and the methodology will work whether their categories match this book's ten or surface others.

What Is Owed

The framework prices what extraction severs from future generations. This is a strong claim, and it deserves a strong philosophical anchor — because critics will push on a particular question: how can extraction wrong people who do not yet exist?

The most rigorous form of this objection is what Derek Parfit named the **non-identity problem**. Different policy choices today determine which specific future people come into existence at all. A world that extracts coal aggressively produces some particular set of future humans living particular lives; a world that extracts coal gently produces a different set of future humans living different lives. The aggressive-extraction children cannot be wronged by the policy that produced them, on a strict person-affecting account, because the alternative for them is non-existence rather than better-existence. This is not an abstruse philosophical puzzle; it is the deepest objection to any framework that prices future-generation costs.

Parfit's response — and the response this framework adopts — is to operate at the level of **outcomes** rather than of specific persons. The moral question is not "did the extraction wrong any particular future person?" but rather "is the future-generation outcome worse than it would have been under different extraction policy?" If the answer is yes, the loss is real, and the accounting matters, regardless of whether the persons who bear the loss could have been the same persons under different policy. This is what philosophers call an *impersonal* moral framework — outcomes are evaluated by their goodness for whoever-the-future-people-turn-out-to-be, not by their goodness for specific identifiable individuals.

The framework's Residual Commons Value calculation operates in exactly this register. RCV does not price what was taken from a particular named miner's grandchild. RCV prices what extraction severs from whatever future generations arise — the commons not available to be lived in, not available to be drawn from, not available to be passed forward. The Foreclosure Bond is the framework's accounting instrument for that severance, paired with the Restitution Bond's accounting for harm already realized. The two-instrument architecture inherits Parfit's distinction between forward-looking impersonal-outcomes evaluation (the Foreclosure Bond's domain) and backward-looking person-affecting evaluation (the Restitution Bond's domain) — both are legitimate moral domains; the framework distinguishes them rather than collapsing them.

The methodological work each instrument requires is also different. Reparations economics has spent half a century developing methodology for already-realized person-affecting harm; environmental-bond economics has spent three decades on prospective impersonal-outcomes loss. Both are real, both are sophisticated. A simpler framework that specified a single Cost Severance accounting object — backward-looking and forward-looking together, undifferentiated — would have failed under stress, because the accounting work each direction requires is structurally different. The two instruments share the same central equation — Cost Severance equals Residual Commons Value minus Bond posting — but they answer different questions on opposite sides of it.

This grounding does not solve the non-identity problem in its philosophical depth. Parfit himself did not solve it; the literature has not solved it; the framework does not pretend to. What the framework does is **operate within** an established moral-philosophy vocabulary for intergenerational status, rather than treating intergenerational obligation as something the reader is asked to accept on faith. The moral standing of future generations is, in Parfit's framework, a defensible philosophical commitment — and the framework's IPG of 33 (the McDowell-coal compression) is the empirical apparatus that prices what is owed under that commitment.

The methodology applied to the Restitution Bond builds on Darity and Mullen's reparations-economics wealth-gap framework, engaged at depth in Chapter 5's introduction of the two-instrument apparatus; Chapter 6's role here is methodological. The framework's damages-already-realized instrument is a specialization of that methodology to extraction-community contexts — portable mechanism, per-context calibration.

A second philosophical grounding the framework operates within — complementary to Parfit's impersonal-outcomes evaluation — is Amartya Sen's capability-and-social-welfare-function framework. Where Parfit answers the question of *whether* future generations have moral standing, Sen answers the question of *how* their standing should be valued: in terms of the capabilities the framework's accounting can preserve or extinguish. The framework's Residual Commons Value calculation is, in Sen's vocabulary, a measurement of the capability set extraction forecloses on whoever-the-future-people-turn-out-to-be. Critics who ask "why give value to future generations?" are asking a Parfit-domain question; the framework's answer is *because they will have capabilities that present extraction will or will not constrain*, and Sen's framework names the capabilities and the valuation methodology that grounds the framework's pricing. The pairing — Parfit grounding the standing, Sen grounding the valuation — is what makes the framework's intergenerational-pricing claim defensible across the moral-philosophy and welfare-economics traditions simultaneously rather than within only one.

Two Kinds of Commons

A reader from the Ostrom tradition will hold a particular question at arm's length when this framework is introduced. The question is whether the Commons Inversion Test, when it filters cost claims, is filtering *the same kind of commons* that Ostrom's *Governing the Commons* describes — and if so, whether the framework is competing with Ostrom-tradition commons-economics or extending it into adjacent territory. The honest answer requires distinguishing two structurally different commons-classes that have, until recently, been treated under one heading.

The first class — call it the coordination commons — is what Ostrom's eight design principles address. A fishery shared by twenty boats. A grazing pasture shared by forty households. An irrigation system shared by a hundred farms. The commons is finite. Multiple users draw from it. The risk Ostrom names is that uncoordinated drawing produces collective ruin — Hardin's "tragedy" — and her contribution was to demonstrate, empirically across hundreds of cases, that communities of shared-stakeholders can selfgovern these commons through specific institutional architectures. The cost the framework would have to price, if it were applied to a coordination commons, is the cost of coordination breakdown. That is not what this framework prices. Coordination breakdown is a cost the existing Ostrom-tradition apparatus already prices, and prices well, where the necessary institutional architecture is in place.

The second class — call it the extraction commons — is what *this* framework addresses. The atmospheric carbon-stability that makes habitable climate. The Appalachian mountain habitat that exists or doesn't. The ancestral land whose dispossession is permanent. The cumulative collective-cognitive-bandwidth that platform attention extraction can deplete or not. Multiple users still draw from these commons. But the actors *capturing* value through extraction are not, in any meaningful sense, shared-stakeholders with the actors *bearing* cost. The mining company is not a member of the McDowell community. The platform is not a member of the user population whose attention it monetizes. The carbon emitter is not a member of the future generation whose habitable climate is being foreclosed. The asymmetry between value-capture and cost-bearing is structural, not contingent. The cost the framework prices is the severance enabled by that asymmetry.

The Commons Inversion Test's two sub-forms make the distinction operational. Commons-Absence Inversion asks whether the cost claim is commons-grounded at all. Commons-Consumption Inversion asks whether it is consumption-of-finite-commons that the cost is grounded in. A coordination commons fails the second test: if the fishery were inexhaustible, the coordination cost would be unchanged — communities would still need to govern access — and so the cost the cost claim names is not severance in this framework's specific sense. An extraction commons passes the second test: if the atmospheric carbon-stability were inexhaustible, extraction would price differently because the cost-bearing party's cost would not exist. The framework operates on extraction commons; Ostrom operates on coordination commons; the two domains are adjacent and the methodologies extend rather than compete with one another. Where a real-world case carries both elements — a forest that is both shared-stakeholder pastoral commons (Ostrom territory) and target of external timber extraction (this framework's territory) — the analyses run in parallel. The Commons Inversion Test screens what this framework can address; Ostrom's eight design principles address what hers can. Each method does work the other can't.

A second break-point Ostrom's framework does not address is the heterogeneous-stakeholder commons: where a community drawing from a shared commons is divided on ethnic, racial, class, or other grounds, and carries conflicting use-preferences for the commons itself. Ostrom's eight design principles presume substantially-aligned stakeholders capable of negotiating governance institutions; when the stakeholders are structurally divided, the design principles do not produce a solution because the problem is not solely coordination-of-mutually-acceptable-use but contestation over what acceptable use is. This framework's apparatus reaches such commons because the Commons Inversion Test operates on the commons-extraction relationship rather than on the stakeholder-coordination relationship; the cost severance the framework prices is the cost of extraction *from* the commons by extracting parties structurally outside the commons-bearing community, and this cost is measurable whether the cost-bearing community is internally aligned or divided. This second break-point connects the framework to the analytical apparatus of stratification economics (Darity, Hamilton, and collaborators), which analyzes structural-extraction operating through hierarchical group-divisions of race, class, and ethnicity. The framework's commons-governance extension applies stratification-economics analysis of intergroup relationships to commons that are simultaneously contested across stakeholder-group lines and being extracted from by structural outsiders.

The Four Gates

The book has carried the framework's discipline implicitly from the earlier chapters. Here, where the formula is specified, the discipline should be named directly. Any cost the framework admits into the residual commons value computation must pass four gates. A cost that fails any one is not included; a cost that passes all four is.

The first gate is the Commons Inversion Test. Faced with a candidate cost — say, the health damages from a coal mine — we ask whether the cost depends on a commons. The procedure is to imagine the commons inverted: either absent altogether (what would it take to replace what was lost?) or unconsumed (what could the commons do if it weren't being drawn down?). When the cost becomes visible only against an inverted commons, the cost is commons-dependent and admitted to RCV. When the cost would persist regardless — when it's intrinsic to the activity rather than to the commons-state — the test rejects it as not commons-grounded.

The test has two operational sub-forms.

Commons-Absence Inversion asks: imagine the commons not existing. Picture extraction stripping a place of habitability the way a coal seam strips a county of its hillsides. Picture the atmospheric oxygen that feels free on Earth simply gone, the way it is in space. The cost the test surfaces is replacement cost — what it would take to recreate or compensate for what was destroyed. McDowell County's habitability commons, demonstrably stripped, generates a Commons-Absence cost: thirteen years of life expectancy, communities that don't refill, ecological remediation work that no one has been priced for.

Commons-Consumption Inversion asks: imagine the commons not being consumed. Picture the lifetime hours that could have gone to learning a language or raising a child if they hadn't been spent commuting two and a half hours each direction. Picture the forest that could have continued to function as carbon sink and watershed if it hadn't been cut. The cost the test surfaces is opportunity cost — what the commons would have done if it weren't being drawn down. Most resource-extraction cases are Commons-Absence; most personal-scale extractions (time, attention, health expenditure) are Commons-Consumption; some, like a mine that destroys a hillside AND consumes lifetime labor, exercise both.

Naming the two sub-forms makes visible what the test was always doing under the abstract phrase "abundance inversion." The grounding object is the commons; the operation is to imagine the commons differently than it currently is.

The two sub-forms run different filters. Commons-Absence Inversion distinguishes commons-grounded claims from private-property externalities (a different framework applies to the latter). Commons-Consumption Inversion distinguishes consumption-of-finite-commons claims from coordination-of-mutually-acceptable-use claims (the Ostrom-tradition framework discussed in the prior section). A single-form Commons Inversion Test would have collapsed these into one test and produced ambiguous results in the most-load-bearing case — an Ostrom-tradition coordination commons where Commons-Absence passes but Commons-Consumption fails. The two-sub-form architecture makes the routing explicit.

The framework's choice of *Commons Inversion Test* as the name for its discovery method is a deliberate one (see Technical Appendix §6 — Commons Inversion Test). Other names were available — *commons reversal test*, *commons recovery test*, *counterfactual commons test* — and each fails to carry the specific structural feature *inversion* names. The test does not reverse anything in the world; reversal implies an action taken on the resource itself, which the test does not perform. The test does not recover anything; recovery implies the resource is restored or made whole, which is not the methodology's claim. *Counterfactual commons test* is structurally accurate but loses the directional specificity the inversion move carries — the test does not run any counterfactual; it runs *one specific* counterfactual, the inversion of the scarcity premise. The mechanism is precise. The test asks, of any proposed cost claim, whether the same physical extraction would still impose that cost if the underlying commons were non-scarce. The world is held fixed in every other respect; only the resource-scarcity dimension is inverted. If the cost survives the inversion — if the same extraction would impose the same cost under non-scarcity — the cost is intrinsic to the activity, not a cost of commons-extraction; it does not enter the framework's accounting. If the cost vanishes under inversion — if non-scarcity would have made the cost go away — the cost is grounded in the commons-extraction relationship and admits to the framework's measurement. The methodology is the David Lewis counterfactual-reasoning tradition specialized to one specific premise-flip. The alternative the framework declined was revealed preference, which reads what current institutional architectures have already priced as the measure of what a commons is worth. Revealed preference reveals what existing institutions have captured, not what the underlying commons costs to extract from. The framework uses revealed preference as one estimation input later, but rejects it as the discrimination gate — the gate has to be independent of the institutional architecture the framework is trying to assess. *Inversion* names what the test does, and what no other available name does as precisely.

The available alternative for filtering cost claims at the gate level was revealed preference — read what polities or markets have already paid as a measure of what a commons is worth. The advantage is empirical: the data exist, the methodology is established. The disadvantage is structural: revealed preference reveals what current institutional architectures have priced, not what the underlying commons is worth. Norway's revealed-preference per-barrel value reveals what Norwegian institutional architecture captured; it does not reveal what the atmospheric externality from combusting Norwegian oil costs the populations bearing that externality. The Commons Inversion Test, by contrast, runs a thought-experiment counterfactual that is independent of current institutional architecture. The framework uses revealed preference as one estimation input within Approach 2, but uses it as estimation, not as the discrimination gate. The two roles are different; the framework specifies them separately because conflating them produces institutionally-captured results.

The second gate is Units Consistency. A candidate must be expressible as dollars per unit of resource per unit of time. This looks like bookkeeping, and it is, but it does real work: it excludes costs specified only in ordinal terms (quality-of-life scores, dignity indices, cultural-importance rankings) until they are converted to dollar-equivalents with transparent methodology. The gate does not forbid the conversion — hedonic pricing, contingent valuation, and revealed-preference methods are all admissible — but the conversion has to be explicit before a term enters a formula whose other terms are already priced in dollars.

The third gate is boundedness. The candidate's contribution to the integral must converge to a finite value under the framework's discount assumptions. A cost that grows without bound as the time horizon extends cannot enter the formula in that form. Most empirically measured costs satisfy this gate automatically; it exists to exclude unbounded theoretical constructions that would otherwise consume any finite residual commons value.

The fourth gate is independence. A candidate must not double-count what an existing term already captures. If two candidates overlap, the framework resolves the overlap by redefining one of them, merging the two into a single broader term, or rejecting the redundant candidate. This is the same discipline the book has applied to the commons categories, now operating at the cost-term level rather than the dimension level.

The Commons Inversion Test alone is necessary for the framework's accounting but not sufficient. A cost claim that passes both Commons-Absence and Commons-Consumption Inversion has demonstrated that it is grounded in consumption of a finite commons; it has not yet demonstrated that its units are consistent (a claim mixing monetary, physical, and qualitative units fails Units Consistency); that its bounds are finite (a claim of "infinite" cost dominates any aggregation); or that it does not double-count across overlapping commons categories (a single harm priced both as Habitability and as Ecosystem inflates the aggregate). The framework specifies all four gates because each does work the others can't; specifying fewer would admit cost claims that are commons-grounded but unit-incoherent, or finite-bounded but double-counted.

All four gates have formal statements, worked examples, and a step-by-step application protocol in the Technical Appendix's Four Gates Admission Apparatus section. The Appendix also derives the formula's general form — RCV as a discounted integral of a sum of cost terms, rather than only the two-term expression introduced above — and shows that the two-term formula is the $n = 2$ special case under the four gates (Formula Generalization section). The same apparatus runs in both temporal directions; the bidirectional applicability of the framework's gates and three methods is articulated in Technical Appendix §5.5. The generalization is what makes the framework extensible. Future analyses can admit cost terms this book does not enumerate, provided each admission passes the gates. The two-term expression remains the formula that produces the numbers in the convergence table. Everything downstream in this chapter and in subsequent analysis sits on top of the generalization.

The Contribution

One more thing the chapter owes the reader before the critics get their hearing. What, specifically, does the residual commons value framework add?

This framework extends two intellectual traditions — Pigouvian externality theory and Ostrom-lineage commons governance — into a setting neither addressed: extraction contexts where commons are drawn down rather than managed-in-place. From Pigou comes the move to internalize externalities through pricing instruments. From Ostrom comes the recognition that commons can be governed sustainably under specific design-principle conditions. The framework adds what neither developed: a measurement methodology (Commons Inversion Test, with two sub-forms) for the commons-extraction relationship; a quantification framework (Residual Commons Value); a pricing instrument (Accountability Bond) sized to the residual commons value rather than to a single externality.

The framework's name for what extraction takes from the commons-side — *Residual Commons Value* — assembles three loaded sub-choices, and each carries a tradition the term means to honor rather than a coinage the framework invented. *Residual* is the resource-economics word for what is left after standard accounting has priced what standard accounting prices. John Hartwick's 1977 rule for sustainable extraction worked the residual side of exhaustible-resource economics: the rents that accrue to extraction are owed forward as residual obligations to future generations. The framework's residual is in that family — it is what remains to be priced after market revenue and standard externality-accounting have done their work, not a substitute for either. *Commons* is the institutional-form word, and the framework uses it in

Elinor Ostrom's specific sense: a commons is a resource governed by — or extracted from — a community of users with a structural relation to the resource. Alternatives were available. *Resource* would have been generic; *public good* would have implied non-rivalrous and non-excludable, which the commons-under-extraction precisely is not; *natural capital* would have collapsed the institutional question into a financial-asset register the framework specifically avoids. *Value* is the third deliberate choice. Mariana Mazzucato distinguishes value created from value extracted; the framework's value is what the commons creates over time for the parties who depend on it — a value extraction takes without paying for. Cost would have inverted the sign and routed through the cost-bearer side of the ledger; wealth would have collapsed into commodity-finance vocabulary; worth would have been too qualitative to dollar-denominate. The three components converge: what is left after standard accounting (Hartwick), priced from the commons-side rather than the extractor-side (Ostrom), in the value-as-created register that names what extraction has been taking without paying (Mazzucato). The term is the convergence.

Three additions, each consequential.

Substitutability-weighting. Standard externality analysis prices damage when damage occurs. A ton of coal emits a certain quantity of CO_2; the CO_2 does damage D; the external cost is D. The calculation is time-of-damage. What this misses is the counterfactual that every future generation faces: if the resource had been left in the ground until a substitute existed, the damage would not have been done at all. Foreclosure cost — the cost of having extracted early, weighted by the probability that no substitute exists at each future time — is a category standard externality analysis does not have. Standard externality pricing charges the extractor for carbon emissions, health damage, and environmental remediation. It does not charge the option value lost when a resource was burned before its substitute existed. The substitutability function is the instrument that translates that counterfactual into a quantity.

The substitutability-weighting addition has a formal precedent worth naming. Hotelling's 1931 exhaustible-resource economics prices the rent rule for private extraction of finite resources but does not price the option-value of leaving resources in situ for future generations. Pigou's externality framework prices the harm extraction does to third parties — pollution, health damage, environmental cost — but does not price that option-value either. Each tradition does what it does well, but neither alone prices what commons-extraction actually costs when both extraction-rent dynamics and ongoing harm-to-others operate simultaneously. The Hotelling Identity is the framework's bridge: per-unit Residual Commons Value minus per-unit standard Hotelling rent equals per-unit Cost Severance — commons-extraction cost is precisely the residual value beyond standard Hotelling rent under honest accounting. It does not replace either tradition; it formalizes what each leaves out.

Integrated architecture. Standard practice treats foreclosure, externalities, community disruption, intergenerational equity, and epistemic loss as separate concerns governed by separate methods. Each has its own literature, its own measurement conventions, its own reviewers. What no single existing method delivers is a formula that integrates them — one expression whose output is the total cost severance for a given extraction, with each component contributing additively under a shared set of gates. The RCV integrand $[1 - S(t \mid t_0)] \cdot U(R, t, Q(t)) + E(R, t)$ unites the foreclosure and externality terms; the general form extends the sum to any additional cost term that passes the framework's gates. This is not a replacement for the component methods. It is a synthesis that lets analysts produce a single number when a single number is what a decision requires — a reclamation bond, a remediation fund, a severance levy — and that lets them show, transparently, how the components contribute.

The Commons Inversion Test. Standard externality analysis prices what the analyst has already decided to price. The category of costs is inherited; the question is only how much. What the book adds is a discovery method. The Commons Inversion Test asks, of any proposed cost, whether the same physical extraction would still impose it if the underlying resource were non-scarce. If yes, the cost is intrinsic to the activity, not a cost of extraction; it does not enter the RCV. If no, the cost is a scarcity-grounded intergenerational claim and belongs in the integrand. This gives analysts a way to discover previously unpriced costs — or to rule out costs that are attention-capturing but not structural — before committing numbers to them. The Commons Inversion Test is the methodology that makes the framework generative: new variables can be added as the test reveals them, each arriving with its own worked reasoning rather than by assertion.

Apply the three additions to the working cases.

McDowell County coal. Under standard Pigouvian accounting, price health, environmental damage, and carbon — a number that, under \$190/ton CO_2, lands around \$449 to \$464 per ton of coal. RCV adds the foreclosure term (what 1960 coal extracted from 2020's renewable alternatives, weighted by substitutability) and lets the Commons Inversion Test probe the community-disruption costs — those Chapter 2 named qualitatively and this chapter's bottom-up walkthrough priced at \$5 to \$15 per ton — that Pigouvian accounting treats as separate policy questions rather than as components of the extraction's true cost. Same case. Same historical data. Larger, more complete number.

Deepwater Horizon. Under pure Pigouvian accounting, the per-barrel external cost is the documented cleanup plus restoration plus lost-ecosystem services — the \$65 billion of Chapter 5's enumeration. Under RCV, that figure is the E term; the foreclosure term adds the lost optionality of extracting later at better substitutability and with better spill-prevention technology; and the Commons Inversion Test surfaces the Gulf-fishery community-disruption cost that standard Pigouvian analysis, organized by pollutant and by medium, does not naturally capture. The two numbers are not in conflict. The RCV number is larger because it integrates more of what was severed.

Libby, Montana. The methodological point lands sharpest here. Standard externality accounting produces a medical-damage figure for asbestos-related illness and a remediation figure for the physical mine site. Under RCV, those costs enter as the E term; the Commons Inversion Test then surfaces the foreclosure cost associated with the destruction of Libby itself as a functioning town — the community-disruption variable — and the downstream lineage labor cost borne by families whose caregiving decades were restructured around asbestos-related illness. Standard Pigouvian analysis has no principled reason to exclude these costs; it simply has no method for including them. RCV has a method.

The three additions are real, and the framework is non-trivial relative to the tradition it grew from. It is not a rejection of externality economics. It is an extension.

What the Chapter Leaves for Later

Two deliberate omissions. Chapter 7 will apply the same framework in a different setting — the Mars colony thought experiment — and use the setting to test whether the framework is Earth-parochial or general. Chapter 8 will then take one extraction, one ton of McDowell County coal, and walk it through the framework's full cost-decomposition — resource-specific rather than resource-general — to show what the computed gap looks like when it is not aggregated, abstracted, or dissolved in policy language.

This chapter has done the mathematics. The next chapter tests whether the mathematics is universal. The chapter after that makes the number concrete for one extraction we have followed from the book's beginning.

A methodological apparatus the chapter has not formally introduced is the framework's decision-rule under deep uncertainty. Standard policy-evaluation methodology is some variant of Cost-Benefit Analysis, which performs well when costs and benefits are symmetrically reversible, expected values are reliable, and the time-horizon is short. Commons-extraction decisions violate all three conditions: irreversible commons-loss cannot be undone, expected values are subject to deep uncertainty over century-scale horizons, and the asymmetry between extracting (mostly irreversible) and not-extracting (revisitable) makes the regret structure asymmetric in a way CBA does not capture. The framework's response is the Asymmetric Regret Rule, which specializes Savage 1951's minimax-regret discipline to commons-extraction decisions where one direction of regret is commons-extinction. The rule does not replace CBA where CBA performs well; it specializes for the case-class where CBA's symmetry assumption fails. The Technical Appendix carries the formal specification (§8, Asymmetric Regret Rule), with empirical evidence from Method 3 (Scarcity-Adjusted Option Value) demonstrating the rule's prescriptions hold across the documented case library. Chapter 9's policy economy applies the rule to specific decisions.

The framework's choice of *Asymmetric Regret Rule* as the name for its decision-rule under deep uncertainty defends at the name level on two distinctions. The *regret* (rather than *risk*) choice reflects that under deep uncertainty over century-scale horizons, the probabilities expected-utility risk would require are not knowable; Savage 1951's minimax-regret discipline, which the framework specializes, operates without them. The *asymmetric* (rather than *one-sided*) choice reflects the specific structural feature commons-extinction produces: one direction's regret is unbounded (commons-extinction is loss of access that cannot be recovered at any cost), and the other direction's regret is bounded (foregone extraction revenue across a fixed time horizon). The framework joins the regret-theory tradition and names what its specialization adds: the asymmetry the irreversibility of commons-extinction produces.

Three methods. Different foundations. Same direction. When markets do not price what honest accounting prices, the difference is not a rounding error. It is the subsidy by which the entire extraction economy runs.

— End of Chapter 6 —